

Road Scholar

(Scheduled Event - 30 Minutes)



Description: Participants will demonstrate skills in map reading, interpretation, and map construction using road maps and topographic maps.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring:

- Pen/Pencil
- Ruler
- Protractor
- Four-function calculator
- Magnifying glass

Materials Provided at Event:

- Virginia road map(s)
- USGS 7.5-minute topographic map(s)
- Blank paper for mapping and scratch work

Safety Requirements: None

The Competition: Participants will answer questions using provided road and topographic maps.

Content topics will include but are not limited to:

1. Map Reading and Interpretation

- Identify and interpret map symbols, keys, and legends.
- Use compass roses to determine direction and bearings.
- Determine coordinates of specific locations.
- Identify boundaries such as counties, states, and other political borders.

2. Distance and Scale

- Calculate distances between locations using map scale.
- Apply scale to real-world distance problems.
- Interpret ratio and bar scales.

3. Topographic Map Skills

- Interpret contour lines to determine elevation and landforms.
- Calculate elevation change between points.
- Determine percent slope between locations.
- Analyze terrain features using elevation data.

4. Road Map Skills

- Determine shortest distance between locations.
- Provide written driving or walking directions using map features.
- Identify locations and features using map legends.

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of responses to selected questions chosen by the Event Supervisor prior to competition.

Resources:

- [Road Scholar](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles